



## GIRRAWEE HIGH SCHOOL

### Task 2

**2014**

# MATHEMATICS

*Time allowed – 90 minutes  
(Plus 5 minutes' reading time)*

### DIRECTIONS TO CANDIDATES

- Attempt ALL 6 questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are on laminated sheet provided.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.



FINAL MARK

**GIRRAWEE HIGH SCHOOL  
MATHEMATICS  
2014**

**HIGHER SCHOOL CERTIFICATE EXAMINATION**

**ANSWERS COVER SHEET**

**Name:** \_\_\_\_\_

| QUESTION     | MARK | H2 | H3 | H4 | H5  | H6  | H7 | H8 | H9  |
|--------------|------|----|----|----|-----|-----|----|----|-----|
| Section 1    | /5   |    |    |    |     |     |    |    | √   |
| Part B<br>Q6 | /10  |    |    |    | √   |     |    |    | √   |
| Q 7          | /14  |    |    |    | √   | √   |    |    | √   |
| Q 8          | /10  |    |    |    | √   | √   |    |    | √   |
| Q 9          | /11  |    |    |    | √   | √   |    |    | √   |
| Q 10         | /22  |    |    |    | √   | √   |    |    | √   |
| Q 11         | /9   |    |    |    | √   | √   |    | √  | √   |
| TOTAL        | /81  |    |    |    | /76 | /66 |    | /9 | /81 |

## **HSC Outcomes**

## **Mathematics**

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts.
- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.

**GIRRAWEE HIGH SCHOOL  
MATHEMATICS**

**YEAR 12 HSC**

**Task 2 2014**

**Time Allowed: 90 minutes**

**INSTRUCTIONS TO STUDENTS**

- Attempt **ALL 6** questions.
- Circle the best response for the questions in **Part A**
- Start each question in **Part B** on a new page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.

**PART A (5 marks)**

**For questions 1-5 circle the best response from the following:**

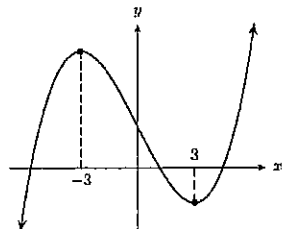
**Question 1.** What are the  $x$ -coordinates of the two stationary points to the curve  $y = 5 + 3x^3 - 2x^4$ ?

- (A)  $x = 0, x = \frac{2}{3}$
- (B)  $x = 0, x = \frac{3}{2}$
- (C)  $x = 0, x = \frac{8}{9}$
- (D)  $x = 0, x = \frac{9}{8}$

**Question 2.** For which values of  $x$  is the curve  $f(x) = -x^2 + 4x + 5$  increasing?

- (A)  $x > 2$                       (B)  $x < 2$                       (C)  $x \geq -2$                       (D)  $x \leq -2$

**Question 3.** From the graph of  $y = f(x)$ , when is  $f'(x)$  negative?

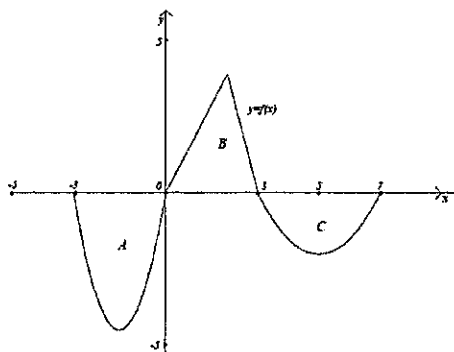


- (A)  $x < -3$  or  $x > 3$                       (B)  $-3 < x < 3$                       (C)  $x \leq -3$  or  $x \geq 3$                       (D)  $-3 \leq x \leq 3$

**Question 4.** What is the value of  $\int_3^5 (x+1)^{-2} dx$ ?

- (A)  $\frac{1}{12}$  (B)  $\frac{1}{24}$  (C)  $\frac{5}{24}$  (D)  $-\frac{5}{24}$

**Question 5.**



The graph on the right shows the curve  $y = f(x)$ . The shaded areas are bounded by  $y = f(x)$  and the  $x$  axis. Shaded area  $A$  is 9 square units, shaded area  $B$  is 6 square units and shaded area  $C$  is 5 square units.

The value of  $\int_{-3}^7 f(x) dx$

- (A) 8 (B) -8 (C) 20 (D) -20

## **PART B**

**QUESTION 6. ( 10 marks)**

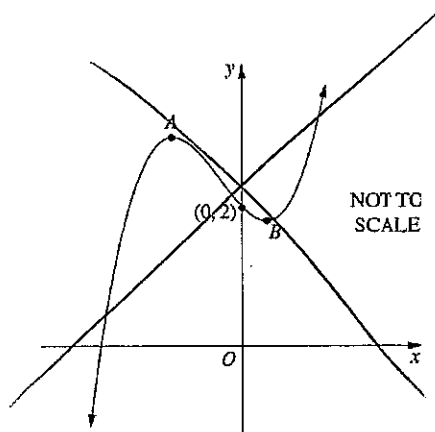
- (a) Show that the curve  $y = 2x - \frac{3}{x-1}$  is increasing for all values of  $x$ . 2
- (b) The curve  $y = x^3 + ax^2 + bx + 5$  has a stationary point at  $(2, -3)$ . Find the values of  $a$  and  $b$ . 3
- (c) If  $y = \frac{x}{x-1}$ , show that  $\frac{2yy'}{x} + y'' = 0$ . 3
- (d) For what values of  $x$  is the curve  $y = 2x^3 + 3x^2 - 12x + 8$  concave up? 2

**QUESTION 7 (14 marks)**

- (a) Show that the curve  $y = x^4 - x + 1$  has no points of inflexion. 3
- (b) The function  $y = x^3 - 3x^2 - 9x + 1$  is defined in the domain  $-4 \leq x \leq 5$ .
- (i) Find the coordinates of any turning points and determine their nature. 4
- (ii) Find the coordinates of any points of inflexion. 2
- (iii) Draw a neat sketch of the curve. 3
- (iv) Determine the minimum value of the function  $y$ , in the domain  $-4 \leq x \leq 5$  2

**Question 8. ( 10 marks)**

(a)

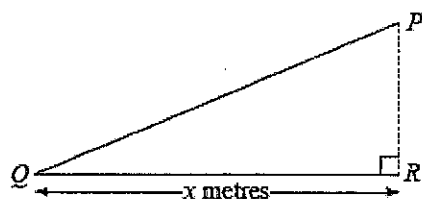


The diagram shows a sketch of the curve  $y = 6x^2 - x^3$ . The curve cuts the x axis at L, and has a local maximum at M and a point of inflection at N.

- (i) Find the coordinates of L. 1
- (ii) Find the coordinates of M. 1
- (iii) Find the coordinates of N. 1

### Question 8 continued

(b)



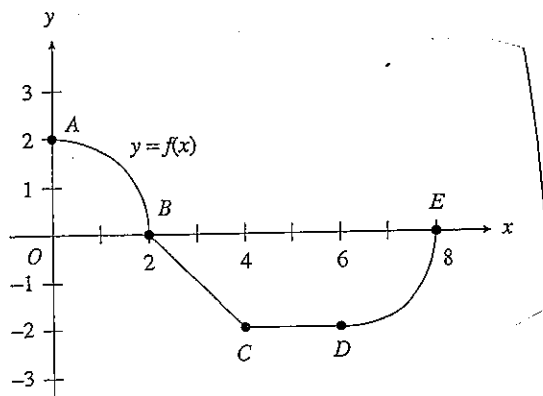
A wire of length 5 metres is to be bent to form the hypotenuse and base of a right-angled triangle PQR, as shown in the diagram. Let the length of the base QR be  $x$  metres.

- (i) What is the length of the hypotenuse PQ in terms of  $x$ ? 1
- (ii) Show that the area of the triangle PQR is  $\frac{1}{2}x\sqrt{25-10x}$  square metres. 2
- (iii) What is the maximum possible area of the triangle? 4

### QUESTION 9. (11marks)

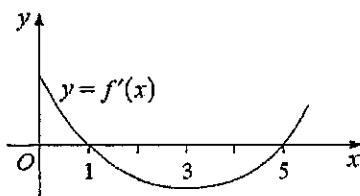
- (a) Find the primitive function of  $\frac{2}{\sqrt{x}} + \frac{1}{\sqrt[3]{x^2}}$ . 3
- (b) Find  $f(x)$ , given that  $f'(x) = 6x^2 - 6x + 5$  and that the curve passes through the point (2,13). 3

(c)



The graph of the function  $f$  consists of a quarter circle AB, a straight line segment BC, a horizontal straight line segment CD, and a quarter circle DE as shown above. Evaluate  $\int_0^8 f(x)dx$ . 3

(d)



The diagram shows the graph of the gradient function of the curve  $y = f(x)$ . For what values of  $x$  does  $f(x)$  have a local minimum? Justify your answer. 2

**Question 10.(22 marks)**

(a) Find each indefinite integral:

(i)  $\int (3x^2 - 2) dx$  **1**

(ii)  $\int (\sqrt{x} - 2)(\sqrt{x} + 2) dx$  **3**

(iii)  $\int \frac{dx}{(2x+3)^2}$  **3**

(iv)  $\int \frac{\sqrt{x}-1}{2\sqrt{x}} dx$  **3**

(b) Evaluate:

(i)  $\int_0^1 (3x^6 + 1) dx$  **2**

(ii)  $\int_0^3 \frac{dx}{(2x-3)^2}$  **3**

(iii)  $\int_2^7 \frac{dx}{\sqrt{x+2}}$  **3**

(iv) (i) Find  $\frac{d}{dx}(x^2 + 1)^5$ . **2**

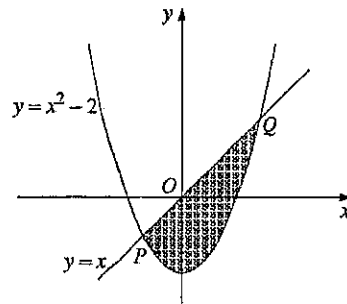
(ii) Hence find  $\int 10x(x^2 + 1)^4 dx$  **2**

**Question 11.(9 marks)**

- (a) Calculate the area enclosed by the curve  $y = 4x^2$ , the  $y$  axis and the lines  $y = 1$ ,  $y = -4$ , lying in the first quadrant.

4

- (b) The diagram shows the graphs of  $y = x^2 - 2$  and  $y = x$ .



- (i) Find the  $x$  values of the points of intersection, P and Q.
- (ii) Calculate the area of the shaded region.

2

3

**End of Examination**



# Year 12 mathematics Task 2 2014 solutions.

## Part A

- ① D      ③ B      ⑤ B  
② B      ④ A

## Part B

### Question 6

a)  $y = 2x - \frac{3}{x-1}, x \neq 1$

$$y' = 2 + 3(x-1)^{-2}$$

$$= 2 + \frac{3}{(x-1)^2}$$

$(x-1)^2 > 0$  for all values of  $x$ .

$\therefore y' > 0$  for all values of  $x$ .

$\therefore$  The curve is increasing for all values of  $x$

(2)

b)  $y = x^3 + ax^2 + bx + 5$

$$y' = 3x^2 + 2ax + b$$

Since  $(2, -3)$  is a stationary point, when  $x = 2, y' = 0$

$$\therefore 0 = 12 + 4a + b$$

$$\therefore 4a + b = -12 \quad \text{--- (1)}$$

Since the curve passes through  $(2, -3)$ ,

in  $y = x^3 + ax^2 + bx + 5$ ,

$$\left. \begin{matrix} x=2 \\ y=-3 \end{matrix} \right\} \quad -3 = 8 + 4a + 2b + 5$$

$$\therefore 4a + 2b = -16 \quad \text{--- (2)}$$

$$4a + b = -12 \quad \text{--- (1)}$$

$$4a + 2b = -16 \quad \text{--- (2)}$$

$$\text{①} - \text{②} \quad -b = 4 \quad \therefore b = -4$$

$$\therefore a = -2 \quad \text{--- (3)}$$

c)  $y = \frac{x}{x-1}$

$$y' = \frac{1(x-1) - x}{(x-1)^2} = \frac{-1}{(x-1)^2}$$

$$y'' = 2(x-1)^{-3} = \frac{2}{(x-1)^3}$$

LHS  $\frac{2yy'}{x} + y'' =$

$$= \frac{2}{x} \times \frac{x}{x-1} \times \frac{-1}{(x-1)^2} + \frac{2}{(x-1)^3}$$

$$= 0 = \text{RHS} \quad \text{--- (3)}$$

d)  $y = 2x^3 + 3x^2 - 12x + 8$

The curve is concave up  $\Rightarrow y'' > 0$

$$y' = 6x^2 + 6x - 12$$

$$y'' = 12x + 6 > 0 \Rightarrow x > -\frac{1}{2}$$

(2)

## Question 7

a)  $y = x^4 - x + 1$

$$y' = 4x^3 - 1$$

$$y'' = 12x^2$$

At points of inflexion,  $y'' = 0$

$$\therefore 12x^2 = 0 \quad \therefore x = 0$$

when  $x = 0$ ,  $y = 1$

$\therefore (0, 1)$  is a possible point of inflexion.

Concavity test:

|       |      |   |     |
|-------|------|---|-----|
| $x$   | -0.1 | 0 | 0.1 |
| $y''$ | +    | 0 | +   |

There is no change of concavity.

$\therefore (0, 1)$  is not a p.o. inflexion.

$\therefore$  The curve has no p.o. inflexion.

(3)

b)  $y = x^3 - 3x^2 - 9x + 1$

$$y' = 3x^2 - 6x - 9$$

At stationary points  $y' = 0$

$$\therefore 3(x^2 - 2x - 3) = 0$$

$$3(x - 3)(x + 1) = 0$$

$\therefore$  stationary points at  $x = -1, 3$

When  $x = -1$ ,  $y = (-1)^3 - 3(-1)^2 - 9(-1) + 1$   
 $= 6$

$x = 3$ ,  $y = -26$

$$y'' = 6x - 6$$

at  $x = -1$ ,  $y'' < 0 \therefore$  maximum

at  $x = 3$ ,  $y'' > 0 \therefore$  minimum

$\therefore$  maximum at  $(-1, 6)$

minimum at  $(3, -26)$  (4)

possible

(i) For points of inflexion

$$y'' = 0 \Rightarrow 6(x - 1) = 0$$

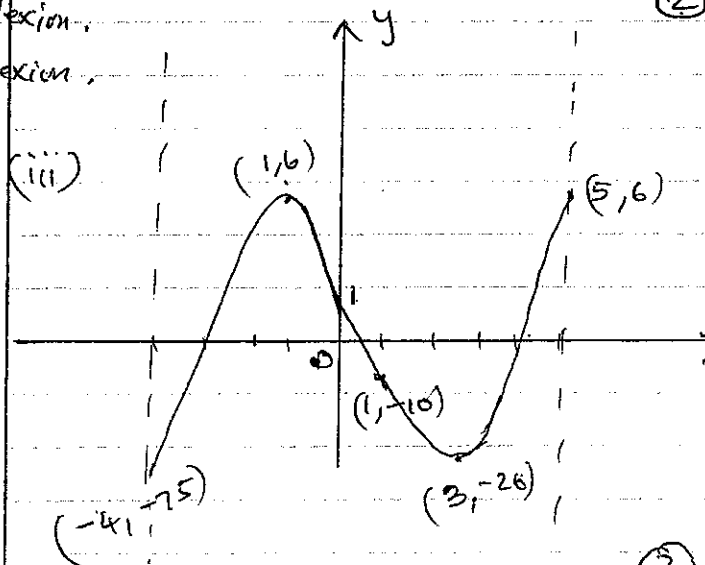
$$\therefore x = 1$$

Concavity test:

|       |    |   |   |
|-------|----|---|---|
| $x$   | -0 | 1 | 2 |
| $y''$ | -  | 0 | + |

$\therefore$  concavity change occurs when  $x = 1$ ,  $y = -10$

$\therefore (1, -10)$  is a point of inflexion (2)



(iii)

(2)

(iv) minimum value = -26

(2)

### Question 8

a) (i)  $y = 6x^2 - x^3$   
 $= x^2(6 - x)$

At L,  $y = 0$

$\therefore x = 0 \text{ or } 6$

$\therefore$  coordinates of L are (6, 0) ①

(ii)  $\frac{dy}{dx} = 12x - 3x^2$   
 $= 3x(4 - x)$

at M,  $\frac{dy}{dx} = 0$

$\therefore 3x(4 - x) = 0$

$\therefore x = 0 \text{ or } 4$

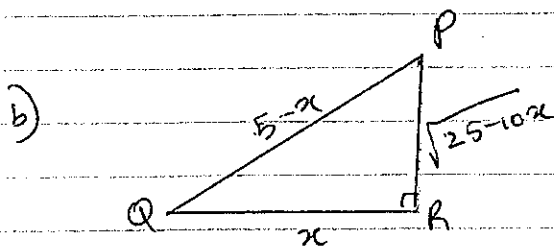
When  $x = 4$ ,  $y = 6 \cdot 4^2 - 4^3$   
 $= 32$

$\therefore N = (4, 32)$  ①

(iii)  $\frac{d^2y}{dx^2} = 12 - 6x = 6(2 - x)$

At N,  $\frac{d^2y}{dx^2} = 0 \Rightarrow x = 2$

$\therefore N = (2, 16)$  ①



(i)  $PQ = (5 - x)m$

(ii)  $PR^2 = (5 - x)^2 - x^2$   
 $= 25 - 10x$

$\therefore PR = \sqrt{25 - 10x}$ , ( $PR > 0$ ) ①

$\therefore$  Area of  $\Delta PQR$

$= \frac{1}{2} \times b \times h$

$= \frac{1}{2} \times x \times \sqrt{25 - 10x}$

$= \frac{1}{2} x \sqrt{25 - 10x} \text{ m}^2$  ②

(iii)  $A = \frac{x}{2} \sqrt{25 - 10x}$

$u = \frac{1}{2}x$   $v = (25 - 10x)^{\frac{1}{2}}$

$u' = \frac{1}{2}$   $v' = \frac{1}{2}(25 - 10x)^{-\frac{1}{2}} \times -10$

$v' = -5(25 - 10x)^{-\frac{1}{2}}$

$v' = \frac{-5}{\sqrt{25 - 10x}}$

$y' = u'v + uv'$

$= \frac{\sqrt{25 - 10x}}{2} - \frac{5x}{2\sqrt{25 - 10x}}$

$= \frac{25 - 10x - 5x}{2\sqrt{25 - 10x}}$

$= \frac{25 - 15x}{2\sqrt{25 - 10x}}$

At stationary points,  $\frac{dA}{dx} = 0$

$\therefore \frac{-15x + 25}{2\sqrt{25 - 10x}} = 0$

$\therefore 15x = 25 \Rightarrow x = \frac{5}{3}$

Concavity test

|                 |   |               |   |
|-----------------|---|---------------|---|
| $x$             | 1 | $\frac{5}{3}$ | 2 |
| $\frac{dA}{dx}$ | + | 0             | - |

$\therefore$  Concavity changes ④

$\therefore$  A maximum occurs at

$x = \frac{5}{3}$

$\therefore A_{\max} = \frac{1}{2} \times \frac{5}{3} \times \sqrt{25 - 10 \times \frac{5}{3}} = \frac{25\sqrt{3}}{18} \text{ m}^2$

### Question 9

$$\begin{aligned} \text{a) } f'(x) &= \frac{2}{\sqrt{x}} + \frac{1}{\sqrt[3]{x^2}} \\ &= 2x^{-\frac{1}{2}} + x^{-\frac{2}{3}} \\ \therefore f(x) &= \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{x^{\frac{1}{3}}}{\frac{1}{3}} + C \\ &= 4x^{\frac{1}{2}} + 3x^{\frac{1}{3}} + C \\ &= 4\sqrt{x} + 3\sqrt[3]{x} + C \end{aligned}$$

### Question 10

d) At  $x=5$ ,  $f'(x)=0$   
 when  $x < 5$ ,  $f'(x) < 0$   
 when  $x > 5$ ,  $f'(x) > 0$   
 $\therefore$  A local minimum at  $x=5$

(2)

$$\begin{aligned} \text{b) } f'(x) &= 6x^2 - 6x + 5 \\ f(x) &= 6 \cdot \frac{x^3}{3} - 6 \cdot \frac{x^2}{2} + 5x + C \\ &= 2x^3 - 3x^2 + 5x + C \end{aligned}$$

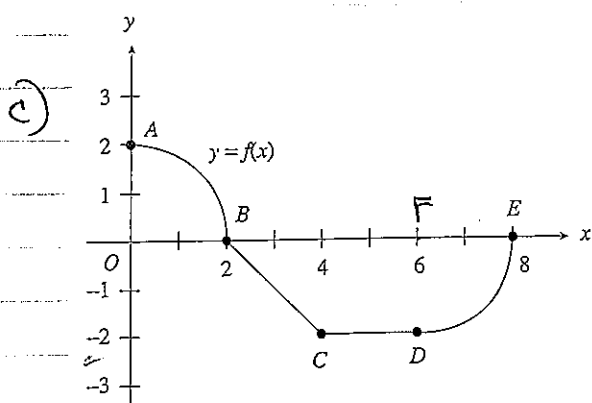
Sub in  $f(2) = 13 \Rightarrow$

$$13 = 16 - 12 + 10 + C$$

$$\therefore C = -1$$

$$\therefore f(x) = 2x^3 - 3x^2 + 5x - 1$$

(3)



Let  $F = (6, 0)$

$$\begin{aligned} \therefore \int_0^8 f(x) dx &= (\text{area } \frac{1}{4} \text{ circle } AB) \\ &\quad - \text{Area of } \frac{1}{4} \text{ circle } DE \\ &\quad - \text{area of trapezium } BCDF \\ &= 0 - \frac{1}{2}(2+4) \cdot 2 = -6 \end{aligned}$$

(3)

$$\text{a) (i) } \int (3x^2 - 2) dx = x^3 - 2x + C$$

(1)

$$\begin{aligned} \text{(ii) } \int (\sqrt{x+2})(\sqrt{x+2}) dx \\ = \int (x+4) dx = \frac{x^2}{2} + 4x + C \end{aligned}$$

(3)

$$\begin{aligned} \text{(iii) } \int \frac{dx}{(2x+3)^2} &= \int (2x+3)^{-2} dx \\ &= \frac{(2x+3)^{-1}}{-1 \times 2} + C \\ &= -\frac{1}{2(2x+3)} + C \end{aligned}$$

(3)

$$\begin{aligned} \text{(iv) } \int \frac{\sqrt{x}-1}{2\sqrt{x}} dx \\ = \int \frac{1}{2} - \frac{1}{2\sqrt{x}} dx \\ = \frac{x}{2} - \frac{1}{2} \cdot \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C \\ = \frac{x}{2} - \sqrt{x} + C \end{aligned}$$

### Question 10

$$\text{a) (i) } \int (3x^2 - 2) dx = x^3 - 2x + C \quad (1)$$

$$\begin{aligned} \text{(ii) } \int (\sqrt{x} - 2)(\sqrt{x} + 2) dx \\ = \int (x - 4) dx = \frac{x^2}{2} - 4x + C \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(iii) } \int \frac{dx}{(2x+3)^2} &= \int (2x+3)^{-2} dx \\ &= -\frac{1}{2(2x+3)} + C \end{aligned}$$

$$\begin{aligned} \text{(iv) } \int \frac{\sqrt{x} - 1}{2\sqrt{x}} dx &= \int \left( \frac{1}{2} - \frac{1}{2\sqrt{x}} \right) dx \\ &= \frac{x}{2} - \frac{2\sqrt{x}^{\frac{1}{2}}}{\frac{1}{2}} + C \\ &= \frac{x}{2} - \sqrt{x} + C \end{aligned} \quad (3)$$

$$\begin{aligned} \text{b) (i) } \int_0^1 (3x^6 + 1) dx &= \left[ \frac{3x^7}{7} + x \right]_0^1 \\ &= 1\frac{3}{7} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(ii) } \int_0^3 \frac{dx}{(2x-3)^2} &= \left[ \frac{(2x-3)^{-1}}{-2} \right]_0^3 \\ &= -\frac{1}{2} \left[ \frac{1}{2x-3} \right]_0^3 = -\frac{1}{2} \left[ \frac{1}{3} + \frac{1}{3} \right] \\ &= -\frac{1}{3} \end{aligned} \quad (3)$$

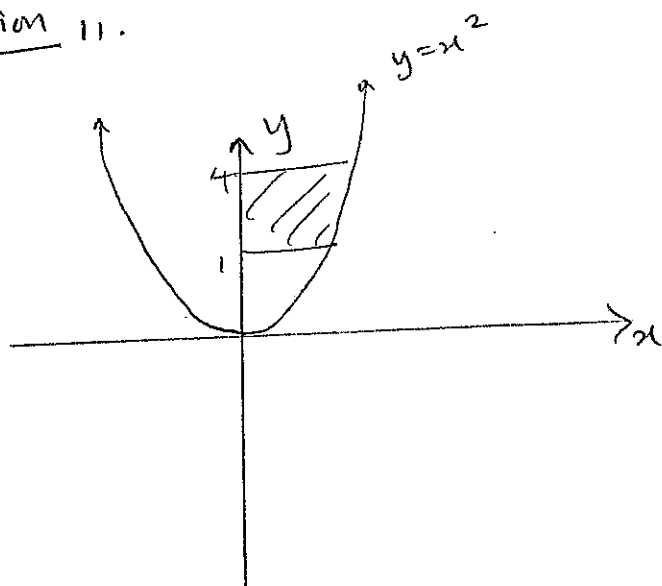
$$\begin{aligned} \text{(ii) } \int_2^3 \frac{dx}{\sqrt{x+2}} &= \int_2^3 (x+2)^{-\frac{1}{2}} dx \\ &= \left[ \frac{(x+2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_2^3 = 2 \left[ \sqrt{x+2} \right]_2^3 \\ &= 2[3 - 2] = 2 \end{aligned} \quad (3)$$

$$\begin{aligned} \text{(iv) (a) } \frac{d}{dx} (x^2 + 1)^5 &= 5(x^2 + 1)^4 \cdot 2x \\ &= 10x(x^2 + 1)^4 \end{aligned}$$

$$\text{(b) } \therefore \int 10x(x^2 + 1)^4 dx = (x^2 + 1)^5 + C$$

Question 11.

a)



$$y = 4x^2 \quad \therefore x = \frac{\sqrt{y}}{2}$$

$$A = \int_1^4 x dy = \int_1^4 \frac{\sqrt{y}}{2} dy$$

$$= \left[ \frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4 = \frac{2}{3} y^{\frac{3}{2}} \Big|_1^4$$

(4)

b) (i) At points of intersection,

$$x^2 - 2 = x$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$\therefore x = -1, 2$$

(2)

(ii) Area of shaded region,

$$= \int_{-1}^2 x - (x^2 - 2) dx$$

$$= \int_{-1}^2 (-x^2 + x + 2) dx$$

$$= \left[ -\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \left[ -\frac{8}{3} + 2 + 4 \right] - \left[ \frac{1}{3} - \frac{1}{2} - 2 \right] = 4\frac{1}{6}$$

### Question 10

$$a) (i) \int (3x^2 - 2) dx = x^3 - 2x + C \quad (1)$$

$$(ii) \int (\sqrt{x} - 2)(\sqrt{x} + 2) dx \\ = \int (x - 4) dx = \frac{x^2}{2} - 4x + C \quad (2)$$

$$(iii) \int \frac{dx}{(2x+3)^2} = \int (2x+3)^{-2} dx \\ = -\frac{1}{2(2x+3)} + C$$

$$(iv) \int \frac{\sqrt{x} - 1}{2\sqrt{x}} = \int \left( \frac{1}{2} - \frac{1}{2\sqrt{x}} \right) dx \\ = \frac{x}{2} - \frac{2\sqrt{x}^{\frac{1}{2}}}{\frac{1}{2}} + C \\ = \frac{x}{2} - \sqrt{x} + C \quad (3)$$

$$b) (i) \int_0^1 (3x^6 + 1) dx = \left[ \frac{3x^7}{7} + x \right]_0^1 \\ = 1\frac{3}{7} \quad (2)$$

$$(ii) \int_0^3 \frac{dx}{(2x-3)^2} = \left[ \frac{(2x-3)^{-1}}{-2} \right]_0^3 \\ = -\frac{1}{2} \left[ \frac{1}{2x-3} \right]_0^3 = -\frac{1}{2} \left[ \frac{1}{3} + \frac{1}{3} \right] \\ = -\frac{1}{3} \quad (3)$$

$$(ii) \int_2^3 \frac{dx}{\sqrt{x+2}} = \int_2^3 (x+2)^{-\frac{1}{2}} dx \\ = \left[ \frac{(x+2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_2^3 = 2 \left[ \sqrt{x+2} \right]_2^3 \\ = 2 [3 - 2] = 2 \quad (3)$$

$$(iv) (a) \frac{d}{dx} (x^2 + 1)^5 = 5(x^2 + 1)^4 \times 2x \\ = 10x(x^2 + 1)^4$$

$$(b) \therefore \int 10x(x^2 + 1)^4 dx = (x^2 + 1)^5 + C$$